B. Tech. (6th Sem)

(Common for CSE, CSE with Specialization in Data Science, CSE with specialization in Cloud Technology & Information Security, CSE with specialization in Full Stack Development, CSE with specialization in AI & Machine Learning)

BCSE-515 (Formal Language & Automata Theory)

L	T	P		
3	1	0	Continuous evaluation	40
			End semester exam	60
			Total marks	100
			Credits	4.0

Course Objectives:

i) To provide fundamental understanding of the core concepts in automata theory and formal languages.

ii) To provide knowledge about how to design automata, regular expressions and context-free grammars accepting or generating a certain language.

iii) To learn the ability to describe the language accepted by automata or generated by a regular expression or a context-free grammar.

iv) To appreciate the power of the Turing Machine, as an abstract automaton, that describes computation, effectively and efficiently.

v) To develop a view on the importance of computational theory.

Unit:-1 **(08 Hours)**

Introduction: Alphabet, languages and grammars, productions and derivation, Chomsky hierarchy of languages.

Regular languages and finite automata: Finite State System, Regular expressions and languages, deterministic finite automata (DFA) and equivalence with regular expressions, nondeterministic finite automata (NFA) and equivalence with DFA, regular grammars and equivalence with finite automata, Closure properties of regular languages, pumping lemma for regular languages, minimization of finite automata.

Unit:-2 **(10 Hours)**

Context-free languages and pushdown automata: Context-free grammars (CFG) and languages (CFL), parse trees, ambiguity in CFG, removal of null & unit production, Chomsky and Greibach normal forms, pumping lemma for context-free languages, closure properties of CFLs, deterministic pushdown automata, nondeterministic pushdown automata (PDA) and equivalence with CFG.

Unit:-3 **(08 Hours)**

Introduction to Machines and Context-sensitive languages: Concept of basic machines, Properties and limitations of FSM, Moore and Mealy Machines, Equivalence of Moore and Mealy Machines, Context-sensitive grammars (CSG) and languages, Linear Bound Automata.

Unit:-4 **(10 Hours)**

Recursive enumerable languages and Turing machines: The basic model for Turing machines (TM), Turing recognizable (recursively enumerable) and Turing-decidable (recursive) languages and their closure properties, variants of Turing machines, nondeterministic TMs and equivalence with deterministic TMs, unrestricted grammars and equivalence with Turing machines, TMs as enumerators, Halting Problem of TM, PCP problem, universal Turing machine.

Course Outcome: After completion of this course, students will be able to:

i) Acquire a fundamental understanding of the core concepts in automata theory and formal languages.

ii) Design automata, regular expressions and context-free grammars accepting or generating a certain language.

iii) Describe the language accepted by automata or generated by a regular expression or a context-free grammar.

iv) Appreciate the power of the Turing Machine, as an abstract automaton, that describes computation, effectively and efficiently.

v) Develop a view on the importance of computational theory.

Instructions for paper setter: All Questions are compulsory. The Question paper is divided in to four sections A, B, C and D. Section A is compulsory and comprises of 12 questions of one mark each, 3 from each unit. The questions shall be asked in such a manner that there are no direct answers including one word answer, fill in the blanks or multiple choice questions. Section B comprises of 4 questions of 2 marks each, one from each unit. Section C Comprises of 4 questions of 4 marks each, one from each unit. Section D Comprises of 4 questions of 6 marks each, one from each unit. There is no overall choice, however internal choice may be provided in section C and D, if paper setter so desires.

Text Books:

1. John Martin, Introduction to Languages and The Theory of Computation, Tata McGraw Hill.
2. John E. Hopcroft, Rajeev Motwani and Jeffrey D. Ullman, Introduction to Automata Theory, Languages, and Computation, Pearson Education Asia.

Reference Books:

1. Harry R. Lewis and Christos H. Papadimitriou, Elements of the Theory of Computation, Pearson Education Asia.
2. Dexter C. Kozen, Automata and Computability, Undergraduate Texts in Computer Science, Springer.
3. Michael Sipser, Introduction to the Theory of Computation, PWS Publishing.

B. Tech. (6thSem) Computer Science & Engineering
BCSE-516 (Mobile Application Development)

L	T	P		
2	0	0	Continuous evaluation	40
			End semester exam	60
			Total marks	100
			Credits	2.0

Course Objectives:

i) To learn about the differences between Android and other mobile development environments.
ii) To understand the rapid prototyping techniques to design and develop sophisticated mobile interfaces.
iii) To understand how Android applications work, their life cycle, manifest, Intents, and using external resources.
iv) To design and develop useful Android applications with compelling user interfaces by using, extending, and creating your own layouts and Views and using Menus.

Unit:-1 **(6 Hours)**

Getting started with Mobility: Mobility landscape, Introduction to mobile devices and mobile platforms, Mobile app development, Overview of Android platform, Setting up the mobile app development environment along with an emulators, A case study on Mobile app development.

Unit:-2 **(8 Hours)**

Building blocks of Mobile Apps: App user interface designing- Mobile UI resources (Layout, UI elements, Draw-able, Menu), Activity-states and life cycle, Interaction amongst activities.

App functionality beyond user interface: Threads, Async task, Services (states and life cycle), Notifications, Broadcast receivers, Telephony and SMS APIs.

Native data handling: on device file I/O, shared preferences, Mobile database such as SQLite, and enterprise data access (via internet/intranet).

Unit:-3 **(5 Hours)**

Sprucing up Mobile Apps: Graphics and animations- custom views, Canvas, Animation APIs, Multimedia- audio/video playback and record, Location awareness and native hardware access (Sensors such as accelerometer and gyroscope).

Unit:-4 **(5 Hours)**

Testing Mobile Apps: Debugging mobile application, White box testing, black box testing and test automation of mobile apps using JUnit for android, Robotium and MonkeyTalk.

Taking apps to Market: Versioning, signing and packaging mobile apps, Distributing apps on mobile market place.

Course Outcomes: After completion of this course, students will be able to:

i) Learn and understand the terminology related to mobile application development.
ii) Understand how Android applications work, their life cycle, manifest, Intents, and using external resources.
iii) Explain the differences between Android and other mobile development environments.
iv) Utilize the power of background services, threads, and notifications.
v) Use Android's communication APIs for SMS, telephony, network management, and internet resources (HTTP).
vi) Build the Android applications.
vii) Secure, tune, package, and deploy Android applications.

Instructions for paper setter: All Questions are compulsory. The Question paper is divided in to four sections A, B, C and D. Section A is compulsory and comprises of 12 questions of one mark each, 3 from each unit. The questions shall be asked in such a manner that there are no direct answers including one word answer, fill in the blanks or multiple choice questions. Section B comprises of 4 questions of 2 marks each, one from each unit. Section C Comprises of 4 questions of 4 marks each, one from each unit. Section D Comprises of 4 questions of 6 marks each, one from each unit. There is no overall choice, however internal choice may be provided in section C and D, if paper setter so desires.

Text Books:

1. Anubhav Pradhan and Anil V. Deshpande, “Composing Mobile Apps: Learn, Explore, Apply Using Android”, Wiley India.
2. Barry Burd, “Android Application Development All-in-one for Dummies”, Wiley, 1st Edition.
3. Carmen Delessio and Lauren Darcey, “Teach Yourself Android application Development in 24 Hours”, SAMS Publication.

Reference Books:

1. Valentino Lee and Heather Schneider, “Mobile Applications: Architecture, Design, and Development”, Hewlett-Packard Professional Books, 2004.

B. Tech. (6th Sem) Computer Science & Engineering
BCSE-516L (Mobile Application Development Lab)

L	T	P		
0	0	4	Continuous evaluation	60
			End semester exam	40
			Total marks	100
			Credits	2.0

Course Objectives:

i) To identify mobile app development challenges and outlines mobile app development approaches and technologies.
ii) To understand mechanisms to respond to device events in an app.
iii) To explore usage of graphics, animation capabilities & implementation of media playback, capture and storage in an app.
iv) To understand strategies and best practices for publishing an app.
v) To implement a live mobile app demonstrating app development from inception to publishing.

List of Practical

1. Write a program to study the life cycle of android activity.
2. Write a program to perform basic arithmetic operations using a calculator.
3. Write a program to study the life cycle of two activities.
4. Write a program to send images and text with the help of implicit intent.
5. Write a program to check the state of the mobile phone using TelephonyManager API.
6. Write a program to send and receive text messages using SMSManager API.
7. Write a program to design menus.
8. Write a program to send notifications.
9. Write a program to play songs using services.
10. Write a program to draw circles, rectangles and other shapes using canvas.
11. Write a program to study different types of animation.
12. Write a program to play audio or video using a built-in app mechanism.

Course Project Description:

Title: Mobile Diagnostic Tool for Android (MDT - Android version)

Backstory/Overview: "MDT - Mobile Diagnostic Tool for Android" is a mobile diagnostic app available for android smartphone and tablets. The basic purpose of this Mobile App is to check your smartphone condition and generate complete phone status report. This mobile app will efficiently test and tell you all about the software and hardware information of your phone. Also, it will make sure that everything is working perfectly after completing tests. This project is divided into four major parts:

Part-1: MDT-UI designing; Part-2: MDT- Mobile Info & Spec; Part-3: MDT-Sensors Test; Part-4: MDT-Data Storage & Memory

Part-1: MDT - UI designing: As a part of this project, you need to create and deliver the UI of MDT - Mobile Diagnostic Tool for Android. As a team member, following points should be in your mind:

- Present the user with only what they need to know (means keep content to minimum and specific)
- A simple design will keep the user at ease with the product. So, keep interface elements to a minimum.
- Screens should be interactive and user-friendly.
- Use the technique of progressive disclosure to show more options.
- Make your app appear fast and responsive

Part-2: MDT - Mobile Info & Spec: In this part, a team member needs to perform the following tasks:

- Check Mobile Info (Device name, model number, IMEI info, etc.)
- Get real software and hardware info (Android version, CPU, RAM, Internal storage, Kernel version, etc.)
- Check battery condition
- Get network signals status
- Check Camera, Sound & Display Screen

Part-3: MDT - Sensors Test: In this part, a team member needs to perform the following tasks:

- Motion Sensors Test (Accelerometer, Gyroscope, Rotation vector sensor, Gravity Sensor)
- Position Sensors Test (Proximity, Orientation, Magnetometer)
- Environment sensors (Light, Ambient temperature, Pressure, Humidity), etc.

Part-4: MDT - Data Storage & Memory: As a team member of this project, you need to perform the following tasks:

- Call Logs
- Internal and External Memory Test
- Report Generation of current status of mobile

Course Outcomes: After the completion of this course, the students will be able to:

i) Install and configure Android application development tools.
ii) Design and develop user Interfaces for the Android platform.
iii) Save state information across important operating system events.
iv) Apply Java/Kotlin programming concepts to Android application development.

B. Tech. (6thSem) Computer Science &Engineering
BCSE-551 (Advanced Java Programming)

L	T	P		
2	0	0	Continuous evaluation	40
			End semester exam	60
			Total marks	100
			Credits	2.0

Course Objectives:

i) To learn about JDBC concepts and Collection Framework in Java.
ii) To understand the use of Java Servlets.
iii) To understand how Java Server Pages are designed.
iv) To acquire knowledge about Java Networking & Remote Method Invocation.

Unit:-1 **(9 Hours)**
Java Database Connectivity: Definition, purpose and architecture of JDBC, JDBC drivers, Steps to connect to a database, DriverManager, Connection, Statement, PreparedStatement, ResultSet, CRUD operations using JDBC.
Collection Framework: Hierarchy of collection framework, ArrayList, LinkList, Stack, Queue (PriorityQueue), Map (HashMaps), Sets (HashSets).

Unit:-2 **(9 Hours)**
Java Servlets: Life cycle of a Servlet, ServletConfig, SevletContext, Reading Servlet parameters, HttpServlet, HttpServletRequest, HttpServletResponse, HttpSession, Handling HTTP request and response, Cookies, Session Tracking, Servlet Listener, Servlet Filters.

Unit:-3 **(9 Hours)**
Java Server Pages: Implicit Objects, JSP Directives, scripting elements, Extracting fields and values, attributes, Sessions in JSP, Cookies, JSP UseBean, Java Standard Tag Library, Expression Language, Custom Tag, Introduction to MVC Pattern.

Unit:-4 **(9 Hours)**
Networking in Java: Networking fundamentals, Client/Server model, Internet addresses, Sockets, networking classes and interfaces using java.net package, TCP/IP and datagram programming, HTTP protocol and URLs.
Remote Method Invocation (RMI): What is RMI, Object Serialization, RMI layer model, Skelton, Stub, java.rmi packages (Remote interface, Naming, RMISecurityManager, RemoteException), java.rmi.registry package (Registry interface, LocateRegistry), java.rmi.server package (RemoteObject, RemoteServer), Parameter passing of non-remote and remote objects, cloning remote objects.

Course Outcomes: After completion of this course, student is able to:

i) Write java applications using JDBC and collections.
ii) Learn and use the concept of java servlets, jsp and beans.
iii) Create java networking applications using java.net and RMI.
iv) Use advance Java programming related concepts in development of various projects/fields.

Instructions for paper setter: All Questions are compulsory. The Question paper is divided in to four sections A, B, C and D. Section A is compulsory and comprises of 12 questions of one mark each, 3 from each unit. The questions shall be asked in such a manner that there are no direct answers including one word answer, fill in the blanks or multiple choice questions. Section B comprises of 4 questions of 2 marks each, one from each unit. Section C Comprises of 4 questions of 4 marks each, one from each unit. Section D Comprises of 4 questions of 6 marks each, one from each unit. There is no overall choice, however internal choice may be provided in section C and D, if paper setter so desires.

Text Books:

1. H.M. Deitel, P.J. Deitel, "Java How to Program", 6th edition Pearson Education, New Delhi
2. Bryan Basham, Kathy Sierra, Bert Bates, "Head First Servlets and JSP".

Reference Books:

1. Cay S. Horstmann, Gray Cornell, "Core Java 2 Volume II-Advanced Features", Sun Microsystems Press, Pearson Education, New Delhi.
2. Herbert Schildt, "Java: The Complete Referenc", 7th Edition, Tata McGraw Hill Publication, Delhi.

B. Tech. (6th Sem) Computer Science & Engineering
BCSE-551L (Advanced Java Programming Lab)

L	T	P		
0	0	2	Continuous evaluation	60
			End semester exam	40
			Total marks	100
			Credits	1.0

Course Objectives:

i) Learn to create java applications using JDBC and collections framework.
ii) Learn to create enterprise applications using java servlets.
iii) Learn and use the java server pages in java enterprise applications.
iv) Learn to create projects based on java networking and RMI.

List of Practical

1. Installation of Eclipse IDE for Enterprise Java and Web Developers.
2. Write a java program to enter employee details in an Employee table with attributes- ID, name and salary using JDBC.
3. Write a java program to search, update and delete an Employee by ID.
4. Create an ArrayList of Employee details- ID, name, and salary and display all employee details from the Employee table using the ArrayList..
5. Create and deploy a java servlet to print welcome on the browser and run it using the apache tomcat server in eclipse.
6. Create and deploy a java servlet to print all the Employee details in the Employee table.
7. Create a java jsp program to enter the new employee details from an HTML form into the Employee table.
8. Create a java jsp program to delete an employee from the employee table.
9. Implement java client-server communication using java networking concepts.
10. Call a remote method to display Employee details from the server using RMI concepts.

Course Project Description:

Create an Event Management enterprise application for MM Engineering College where students can register themselves for an upcoming event and results can be published after the event. New events can be created and published by coordinators. Use all the concepts learned in the above practicals.

Course Outcome: After completion of this course, student is able to:

i) Setup client server communication.
ii) Develop enterprise application with collection framework.
iii) Learn to develop JSP based application.
iv) Use the concept of java servlets to develop applications.

Open Elective
BCIV-180: Disaster Management
(Common to All Branches)

L	T	P		Continuous Evaluation	40
3	0	0		End Semester Exam	60
				Total Marks	100
				Credits	3.0

Course Objectives: *The objectives of this course are to:*
1. Discuss the importance of planning for disasters.
2. To study inter-relationship between Disaster and Development
3. Discuss (overview of) policies and the roles of the local, state, and central government.
4. Discuss various components of disaster relief.

UNIT-I

Introduction to Disasters: Concepts and definitions (Disaster, Hazard, Vulnerability, and Resilience Risks) Disasters: Classification Causes, Impacts (including social, economic, political, environmental, health, psychosocial etc.) Differential impacts – in terms of caste, class, gender, age, location, disability Global trends in disasters urban disasters, pandemics, complex emergencies, Climate change.

UNIT-II

Approaches to Disaster Risk reduction: Disaster cycle – its analysis, Phases, culture of safety, prevention, mitigation and preparedness, community based DRR, structural-non-structural measures, roles and responsibilities of community, Panchayati Raj institutions/Urban local bodies (PRIs/ULBs), states, Centre and other stake-holders.

UNIT-III

Inter-relationship between Disaster and Development: Factors affecting Vulnerabilities, differential impacts, impact of Development projects such as dams, embankments and changes in Land use etc. climate change adaption. Relevance of indigenous knowledge, appropriate technology and local resources.

UNIT-IV

Disaster Risk Management in India: Hazard and vulnerability profile of India, Components of Disaster Relief: Water, Food, Sanitation, Shelter, Health, and Waste Management, Institutional arrangements (Mitigation, Response and Preparedness, DM Act and Policy, other related policies, plans, programmes and legislation)

Project Work: (Field Work and Case studies): The project/fieldwork is meant for students to understand vulnerabilities and to work on reducing disaster risks and to build a culture of safety. Projects must be conceived base on the geographic location and hazard profile of the region where the college is located. A few ideas or suggestions are discussed below: Several governmental initiatives require urban local bodies (ULBs and Panchayati Raj institutions (PRIs) to be proactive in preparing DM plans and community based disaster preparedness plans. Information on these would be available with the district collector or Municipal Corporations. The scope for students to collaborate on these initiatives in immense.

Course Outcomes: *On completion of this course successfully, a candidate will be able to:*
1. Comprehend the key perspectives of disasters, hazards and the related concepts.
2. Prevent, mitigate, develop preparedness, response and plan recovery for disaster risk reduction.
3. Discuss the roles and responsibility of Govt. and private bodies for disaster management.
4. Prepare a comprehensive hazard and vulnerability profile of India.

Note for Paper-setter: The Question paper is divided in to three sections A and B. Section A comprises of 12 questions of one mark each, 3 from each unit. The questions shall be asked in such a manner that there are no direct answers including one word answer, fill in the blanks or multiple choice questions. The section B comprises 4 questions of 12 marks each, one from each unit. Each question shall have two alternatives, out of which student will be required to attempt one

Text/Reference Books:

01. Andharia J. Vulnerability in Disaster Discourse, JTCDM, Tata Institute of Social Sciences Working paper no. 8, 2008
02. Blaikle, P, Cannon T, Davis I, Wisner B 1997. At Risk Natural Hazards, Peoples, Vulnerability and Disasters, Routledge.
03. Coppola P Damon, 2007. Introduction to international Disaster Management
04. Document on World Summit on Sustainable Development 2002
05. Government of India, 2009. National Disaster Management Policy.

Department of Mathematics & Humanities
B.Tech
Indian Constitution
(BHUM-116)

w.e.f Session: 2020-2021

Maximum Marks: 60+40=100
Written Examination-60
Internal Assessment-40
Max. Time - 3 hrs.

Important Note:

(i) The whole syllabi shall be divided into four units I to IV.
(ii) The question paper shall carry two Parts (Part 'A' and Part 'B').
(iii) Part 'A' shall comprise one Compulsory Question of 12 marks containing four short-answer questions spread over all the four units of the syllabi, each question carrying 3 marks.
(iv) There shall be eight questions in Part 'B' with two questions from each of the four units. The candidates shall be required to attempt four questions from Part 'B' selecting one question from each unit. Each question shall carry 12 marks.

UNIT-I

Salient features of Indian Constitution, Preamble, Fundamental Rights: Right to Equality, Fundamental Freedoms to life and personal liberty, Right against exploitation, Freedom of Religion, Cultural & Educational Rights, Right to constitutional remedies, Directive Principles of State Policy, Fundamental Duties

UNIT II

Union Executive

Form of Union Executive- President- Election, Impeachment, Privileges, Immunities; Constitutional Position and Powers of the President; Prime Minister and Council of Ministers; Executive Functions; Attorney General

State Executive

Powers and Constitutional Position of Governor; Chief Minister & Council of Minister; Advocate General

UNIT III

Union Parliament & State Legislatures

Composition, Qualifications, Disqualifications, Sessions
Speaker and Chairman
Functions of the Legislature
Parliamentary Privileges
Different types of Bills
Election Commission

UNIT IV

Federal System in India
Services and Public Service Commission
Emergency Provisions
Amendment of the Constitution; Basic Feature Theory

Suggested Books:

1. G. Austin : Indian Constitution: Cornerstone of a Nation
4. D.D. Basu : Shorter Constitution of India
5. D.D. Basu : Introduction to Constitution
6. M.P. Jain : Constitution of India
7. V.N. Shukla : Constitution of India
8. Narender Kumar : Constitutional Law of India

B. Tech. (6th Sem)

(Common for CSE, CSE with Specialization in Data Science, CSE with specialization in Cloud Technology & Information Security, CSE with specialization in Full Stack Development, CSE with specialization in AI & Machine Learning)

PR-IV (Project-IV)

L	T	P		
0	0	4	Continuous evaluation	60
			End semester exam	40
			Total marks	100
			Credits	2.0

Course Objectives:

i) To learn about various phases of software development life cycle.
ii) To learn about how to provide software solution for real life problems.
iii) To learn about coding and testing of solutions.
iv) To learn about report writing concepts.

The students are required to develop a project during semester and project work evaluation will be entirely based upon project evaluation rubrics as given below:

	Marks Distribution & Criteria	Excellent	Very Good	Good	Poor
		5	4	3	2
Synopsis	Novelty of the Problem Definition / Motivation **(Max Marks 5)**	The given problem definition is novel in nature (81% to100%)	The given problem definition is somewhat novel in nature. (61% to 80%)	The given problem definition is novel in nature to some extent. (41% to 60%)	The given problem definition is not novel in nature. (<40%)
	Objectives / Modeling / Feasibility. Requirement and Scope of the Project **(Max Marks 5)**	Provide a clear purpose of the idea and evidence that supports the project concept. Software Development Process Model is used (Waterfall, Incremental, RAD etc.) (81% to 100%)	Somewhat clear purpose of the idea and evidence that supports the project concept And software Development Process Model is used (Waterfall, Incremental, RAD etc.) (61% to 80%)	Attempts to define purpose of the idea and evidence that support the project concept and Attempts to use software Development Process Model (Water fall, Incremental, RAD etc.) (41% to 60%)	Does not clearly define the purposeof the idea and evidence that supports the project concept. Does not clearly used software Development Process Model (Waterfall, Incremental, RAD etc.) (<40%)
Progress 1	Coding and Implementation **(Max Marks 5)**	5	4	3	2
		Code is correctly implemented. (81% to 100%)	Somewhat Code is correctly implemented. (61% to 80%)	Attempts to code and implement correctly (41% to 60%)	Code is not completely and correctly implemented <40%
	Unit Testing **(Max Marks 5)**	5	4	3	2
		Student delivered presentation covering Validation and Testing. (81% to 100%)	Student delivered somewhat presentation covering Validation and Testing.(61% to 80%)	Student delivered presentation covering few of Validation and Testing. (41% to 60%)	Student not delivered presentation covering Validation and Testing. <40%
		5	4	3	2
	Understanding, Individual Involvement/ Contribution in Project. **(Max Marks 5)**	**Ability** to work within the team. **Willingness** to perform tasks. **Punctuality** On time for team meetings. **Reliability** Perform tasks within time. **Creativity** Provide meaningful insight to the project team. (81% to 100%)	**Somewhat Ability** to work within the team. **Somewhat Willingness** to perform tasks. **Somewhat Punctuality -** On time for team meetings. **Somewhat Reliability** Perform tasks within time. **Somewhat Creativity** Provide meaningful insight to project team. (61% to 80%)	**Attempted Ability** to work within the team. **Attempted Willingness** to perform tasks. **Attempted Punctuality -** On time for team meetings. **Attempted Reliability Dependability** Perform tasks within time. **Attempted Creativity** Provide meaningful insight to project team. (41% to 60%)	**No Ability** to work within the team. **No Willingness** to perform tasks. **No Punctuality-**On time for team meetings. **No Reliability Dependability** Perform tasks within time. **No Creativity** Provide meaningful insight to project team. <40%

Progress 2	Coding and Implementation **(Max Marks 5)**	**5** Code is correctly implemented (81% to 100%)	**4** Somewhat Code is correctly implemented. (61% to 80%)	**3** Attempts to code and implement correctly (41% to 60%)	**2** Code is not completely and correctly implemented <40%
	Testing **(Max Marks 5)**	**5** Student delivered presentation covering Validation and Testing. (81% to 100%)	**4** Student delivered somewhat presentation covering Validation and Testing. (61% to 80%)	**3** Student delivered presentation covering few of Validation and Testing. (41% to 60%)	**2** Student not delivered presentation covering Validation and Testing. <40%
	Understanding, Individual Involvement/ Contribution in Project. **(Max Marks 5)**	**5** **Ability** to work within the team. **Willingness** to perform tasks. **Punctuality** On time for team meetings. **Reliability** Perform tasks within time. **Creativity** Provide meaningful insight to the project team. (81% to 100%)	**4** **Somewhat Ability** to work within the team. **Somewhat Willingness** to perform tasks. **Somewhat Punctuality-** On time for team meetings. **Somewhat Reliability** Perform tasks within time. **Somewhat Creativity** Provide meaning full insight to project team. (61% to 80%)	**3** **Attempted Ability** to work within the team. **Attempted Willingness** to perform tasks. **Attempted Punctuality-** On time for team meetings. **Attempted Reliability Dependability** Perform tasks within time. **Attempted Creativity** Provide meaningful insight to project team. (41% to 60%)	**2** **No Ability** to work within the team. **No Willingness** to perform tasks. **No Punctuality-**On time for team meetings. **No Reliability Dependability** Perform tasks within time. **No Creativity** Provide meaningful insight to project team. <40%
Final Progress	Final Coding and Implementation **(Max Marks 5)**	**5** Code completely and correctly implemented. The design and language used for coding is Correctly chosen. (81% to 100%)	**4** Somewhat Code completely and correctly implements the design and language used for coding is correctly chosen. (61% to 80%)	**3** Attempts to code completely and correctly implement the design and language used for coding is Properly chosen. (41% to 60%)	**2** Code not completely and correctly implement the design and language used for coding is not Correctly chosen. <40%
	Testing **(Max Marks 5)**	**5** Student delivered presentation covering Validation and Testing. (81% to 100%)	**4** Student delivered somewhat presentation covering Validation and Testing (61% to 80%)	**3** Student delivered presentation covering few of Validation and Testing (41% to 60%)	**2** Student not delivered presentation covering Validation and Testing. <40%
	Demonstration cum Presentation **(Max Marks 5)**	**5** Presentation is well organized and reflects logical order. (81% to 100%)	**4** Some of the Presentation does not reflect logical order. (61% to 80%)	**3** Most of the Presentation does not reflect logical order. (41% to 60%)	**2** Presentation does not reflect logical order. <40%
Project Report	Documentation & Report **(Max Marks 5)**	**5** Report as per format provided. The project may be carried for Participation in various contests, published, patented and applied for copyright (81% to 100%)	**4** Report is provided somewhat as per format. The project may be carried for Participation in various contests, published, patented and applied for copyright (61% to 80%)	**3** Report is provided as per format to some extent. Attempted Participation in various contests, Publications, Copyright, Patent (41% to 60%)	**2** Report is not as per the format provided. Poor Participation In various contests, Publications, Copyright, Patent <40%

Course Outcomes:

i) Able to identify software solution corresponding to real life problems.
ii) Able to code software solution.
iii) Able to test software solution.
iv) Able to write reports.